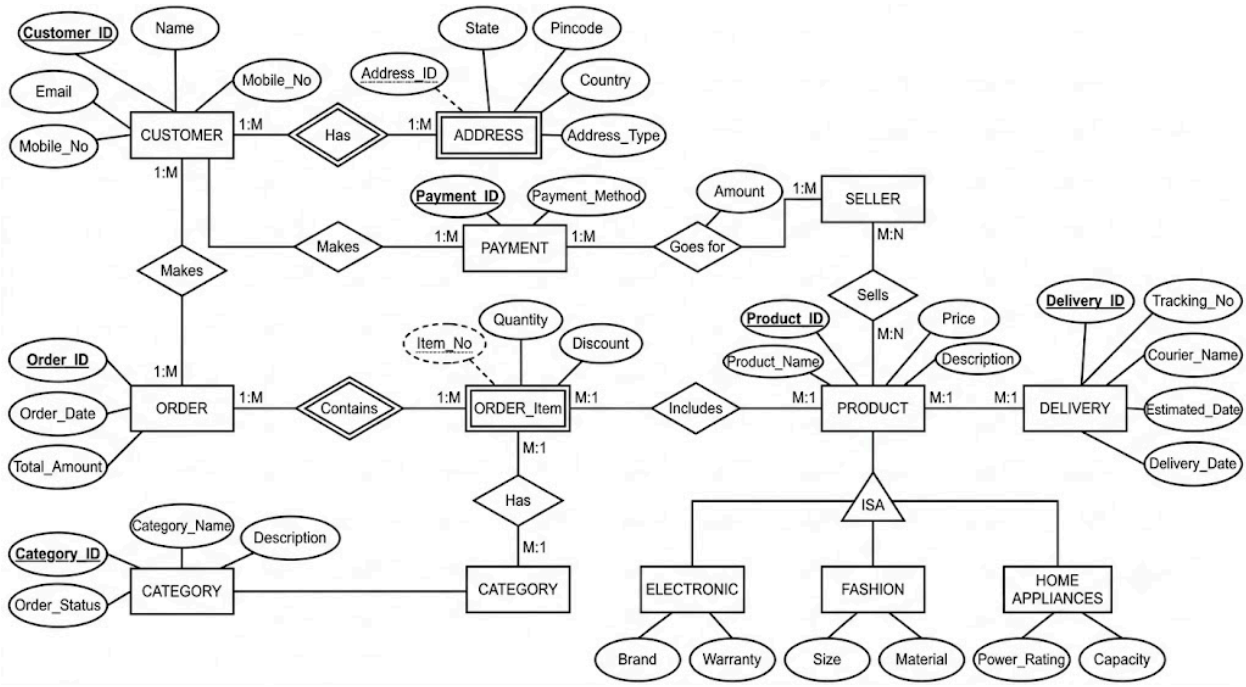
# EXPERIMENT 1

**AIM -** Design an ER diagram for an Indian e-commerce platform with entities Customer, Product, Order, OrderItem, Seller, Category, Payment, Delivery, Address. Specify primary keys, composite and multi-valued attributes, weak entities, specialization of Product, and participation constraints.

## ER DIAGRAM -




## OUTPUT -

- **Successful ER to Relational Mapping:** The ER diagram was successfully translated into a working relational schema. Strong entities were mapped to independent tables, while weak entities (Address, Order_Item) were effectively implemented using composite primary keys that include the parent entity's foreign key.
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- **Specialization (ISA) Implementation:** The inheritance relationship for the PRODUCT entity was successfully modeled using disjoint subtypes. Sub-tables (Electronic, Fashion, Home_Appliances) correctly utilized product_id as both a primary key and a foreign key to link back to the parent Product table.
- **Prevention of Orphan Records (Insertion):** The database successfully prevented the insertion of child records that referenced non-existent parent records, throwing Error 1452 (Foreign Key Constraint Fails) when invalid order_id or product_id values were tested.

- **ON DELETE CASCADE Verification:** It was observed that deleting a parent record (e.g., a Customer) automatically and safely removed all associated child records (such as their Orders and Address) without leaving orphaned data behind.

- **ON DELETE SET NULL Verification:** Deleting a Category record demonstrated that the associated Order_Item rows were retained in the database, but their category_id foreign key was automatically updated to NULL, preserving the order history while removing the category link.

- **Entity and Domain Integrity Checked:** The implementation of UNIQUE and NOT NULL constraints successfully blocked the insertion of duplicate values (e.g., duplicate emails throwing Error 1062) and incomplete records (e.g., missing dates throwing Error 1364).